TECHNICAL DOCUMENTATION

APPLIED AI · RAG

2026 EDITION

Rehabilitation — façades, roofs and existing buildings

REFERENCE DOCUMENTATION FOR THE AI ASSISTANT

RIANSARES 4R

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riansares.com **athagon.tech**

Rehabilitation — façades, roofs and existing buildings

Should all render be stripped from an old façade?

Only after checking adhesion, compatibility, salts, damp and substrate condition. Indiscriminate removal may destroy recoverable finishes and increase cost without resolving the source.

Can cement mortar be applied to an old masonry wall?

Compatibility depends on the wall, moisture, strength, breathability and building conditions. In traditional systems, an excessively rigid or impermeable layer may cause problems; specific assessment is required.

What should be checked on a roof before replacing tiles?

Water entry points, gradients, junctions, gutters, supports, fixings, insulation and safe access. Repositioning a few tiles may not resolve a leak at a junction or valley.

What should be done if deteriorated roof timber is discovered?

Stop work that may compromise stability, assess the cause of deterioration and request review by a competent professional where structural doubt exists. Do not replace a load-bearing member with an improvised substitute.

Can a façade be insulated without changing its appearance?

Internal and external strategies have different effects on moisture, thermal bridges and usable area. Protected buildings may have heritage restrictions; assess the whole before choosing.

How should new openings in an old wall be approached?

Any potentially structural modification requires assessment of loads, material, bearings and relevant authorisations. Do not treat an opening’s size alone as proof of safety.

Is ventilating a damp basement all year advisable?

Ventilation must address the cause; introducing warm, humid outdoor air may cause condensation on cold surfaces. Combine diagnosis of water, temperature, humidity and ventilation.

Which symptoms require structural inspection?

Progressive cracks, out-of-plumb walls, subsidence, doors suddenly failing to fit or significant deformation. Do not diagnose structural stability remotely.

Can just part of a façade be rehabilitated?

It may be feasible if continuity, watertightness, compatibility and agreed appearance are maintained. Edges and junctions must be checked to avoid transferring defects to adjoining areas.

What should be documented before demolishing a partition?

Location, thickness, continuity between floors, services and potential function. Even if it appears lightweight, confirm its nature and local requirements before removal.

Case RE-11 — Façade with falling render

Hypothetical assessment case, not completed work or a service committed to by Riansares 4R.

Situation: Old render is falling near a pedestrian route.

Checks: Falling-material risk, stability and substrate damage.

Possible scope: Protect the area and repair according to diagnosis.

Decision criterion: Prioritise third-party safety before appearance.

Evidence to retain: Photographs, report and checks.

Case RE-12 — Roof dripping in one corner

Hypothetical assessment case, not completed work or a service committed to by Riansares 4R.

Situation: Localised water entry during wind-driven rain.

Checks: Gutter, valley, flashing and substrate.

Possible scope: Repair and verify the junction.

Decision criterion: Do not quote for a complete roof without establishing cause and extent.

Evidence to retain: Plans and detailed photographs.

Industry references and scope

- CTE_HS.
- CTE_HE.
- HSE_HEIGHT.

The references provide general context and do not establish that Riansares 4R has carried out work. Technical requirements must be verified for the applicable type of intervention and jurisdiction.